

MICHAEL PEÑA

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SUMMARY

Master's-trained statistician with experience conducting quantitative research, statistical analysis, and data-driven decision support. Skilled in data collection, data manipulation, statistical modeling, exploratory data analysis, reporting, and presentation of findings. Experienced in preparing technical reports, communicating analytical results to technical and non-technical audiences, and supporting research initiatives through project design and execution. Proficient in R, Excel, SQL, and data visualization techniques. Strong background in transforming complex datasets into meaningful information for stakeholders.

EDUCATION & CERTIFICATES

California State University, Fullerton M.S., Statistics	May 2025
California Polytechnic State University, Pomona B.S., Applied Mathematics and Statistics	May 2023
IBM via edX.org Certificate of Completion, SQL for Data Science	December 2023
IBM via edX.org Certificate of Completion, Linux Commands & Shell Scripting Basics	June 2023

SKILLS

Programming	R, Python, SQL, Bash, VBA
Software	Microsoft Excel, Microsoft 365 Suite, Microsoft Copilot, STATA, R Shiny
Analytical Methods	Quantitative Research Methodologies, Statistical Inference, Exploratory Data Analysis, Descriptive Summary Statistics, Statistical Significance Testing, Time Series Analysis, Survival Analysis, Bayesian Data Analysis, Predictive Modeling, Data Visualization, Data Collection and Validation
Research & Reporting	Technical Report Preparation, Data Interpretation, Stakeholder Presentations, Research Documentation

PUBLICATIONS

Lanza, M.B., Salehpour, S., Addison, O., et al. (2026). A pilot study on aging-related effects on step performance: The role of muscle quality, size, and strength. *Scientific Reports*. <https://doi.org/10.1038/s41598-026-57916-1>

Contributing author and statistical consultant; conducted data preparation, exploratory data analysis, principal component analysis, statistical modeling, and manuscript development for a peer-reviewed study on aging and fall risk.

EXPERIENCE

Tutor Expert Mount San Antonio College, Walnut, California	November 2023 – Present
• Provided assistance to 100+ students in manipulating and formatting datasets to remove inconsistencies, and assisted in data interpretation as part of their coursework and projects • Supported 100+ students in analyzing quantitative data, interpreting results, and providing feedback for written reports • Guided students through statistical analysis methodologies, database concepts, and data visualization techniques using appropriate analytical software • Instructed 100+ students in drafting, reviewing, and editing technical reports, maintaining standards for accuracy and clarity so findings were accurately conveyed	
Statistical Consultant University of Maryland School of Medicine (Remote)	February 2025 – June 2026
• Served as lead statistical consultant on a multidisciplinary study of fall risk in older adults, translating raw laboratory data into a published peer-reviewed research report and data-driven recommendations for future study design • Coordinated development of the statistical analysis plan with a multidisciplinary research team of 3 graduate researchers and 5 clinical researchers for a pilot study assessing fall risk in older adults, monitoring data quality and project progress through completion • Conducted exploratory data analysis, descriptive summary statistics, and statistical significance testing on high-dimensional kinematic and ultrasound imaging data to identify meaningful information and support	

research objectives

- Designed, developed, tested, and implemented statistical models including a k-nearest neighbor classification model in powerful statistical software for quantitative data interpretation
- Formulated and modified analysis code and database queries in R and SQL and installed and maintained statistical software environments to support data validation and reproducible research workflows
- Prepared technical reports and co-authored the statistical analysis and discussion sections of a peer-reviewed publication in *Scientific Reports*
- Provided reports and presented findings during stakeholder briefings, delivering data-driven recommendations to guide project direction

Intern, New Student Programs & Family Engagement | Cal Poly Pomona, California *July 2023 – September 2023*

- Created clear and detailed itineraries and contact sheets in Microsoft Excel for multiple university orientations, ensuring smooth event coordination with attention to accuracy
- Prepared and printed orientation materials and delivered impactful presentations to large groups of family members and prospective students, educating them about the university and its resources

Student Assistant & Test Proctor, Disability Resource Center | Cal Poly Pomona, California *August 2021 – May 2023*

- Applied effective interpersonal communication rules and guidelines with hundreds of students who had learning disabilities
- Created several in-depth operating procedures for future student assistants to follow, maintaining standards for exam administration, documentation, and record keeping

PERSONAL & ACADEMIC PROJECTS

Blood Types Distribution | Personal Project

October 2025

- Designed and implemented a complex hierarchical Bayesian sampling model of human blood type distribution across 17 Mexican states, informed by academic research literature

Electromyogram Prediction | Academic Project, Time Series Analysis

July 2024

- Developed, tested, and presented a model to predict the type of active muscle, trained on 2.4+ million observed biostatistical muscle electromyogram datapoints

Transit Oriented Community Housing Costs | Academic Project, Statistical Learning

Fall 2024

- Conducted statistical survival analysis on 300+ datapoints to demonstrate the relationship between delayed construction and affordable housing, modifying and improving methods from previous academic literature

Building 8, Elevator Wait-Time Prediction | Academic Project, Statistical Consulting

Spring 2023

- Designed a statistical model that accurately predicts the wait time of an elevator using on-site quantitative data collection

RELATED COURSEWORK

Statistical Inference and Probability

Fundamentals of probability and mathematical intuition for statistics and statistical analysis

Statistical Computation

Application of algorithms and simulations using software programming

Categorical Data Analysis

Training of statistical models to accurately predict qualitative outcomes

Time Series Analysis

Application of training time-dependent predictive models

Statistical Learning

Machine learning and predictive training aspects of statistics

Bayesian Data Analysis

Bayesian statistical methodologies with emphasis on biological and life science applications

RESEARCH SUMMARY

As part of my Master's Statistical Consulting Capstone, I worked under clinical researcher Dr. Marcel Lanza on a pilot study examining fall risk in older adults, a project now completed and published in *Scientific Reports* (2026). I conducted exploratory data analysis and data imputation, built visualizations for trend and risk-factor analysis, applied Principal Component Analysis (PCA) for data reduction, and applied a multiple response multiple regression model to evaluate relationships among age, muscle characteristics, and stepping performance. Although the analysis did not identify statistically significant relationships among the variables of interest, the findings provided valuable insight into the limitations of the available sample and informed recommendations for future research. The results indicated that additional observations would be necessary to more fully evaluate the relationship between age, muscle structure, and fall-risk related performance measures.